

## Lab 8 – Diego Licardié 1251825

```
semana_8 > lab8 > mensaje_DiegoLicardie.json > archivo
1 {
2   "nombre": "Diego Licardie",
3   "curso": "Manejo e Implementación de Archivos",
4   "carne": "1251825",
5   "archivo": "mensaje_Diego_Licardie.txt"
6 }
```

```
PS C:\url-mia-2026> git --version
git version 2.55.0.windows.2
PS C:\url-mia-2026> ssh -T git@github.com
Hi Diego35647890! You've successfully authenticated, but GitHub does not provide shell access.
PS C:\url-mia-2026> git checkout -b lab-diego
Switched to a new branch 'lab-diego'
PS C:\url-mia-2026> code mensaje_DiegoLicardie.json
PS C:\url-mia-2026> openssl version
openssl : The term 'openssl' is not recognized as the name of a
cmdlet, function, script file, or operable program. Check the spelling
of the name, or if a path was included, verify that the path is
correct and try again.
At line:1 char:1
+ openssl version
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (openssl:String) [], Com
mandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\url-mia-2026> winget install --id ShiningLight.OpenSSL.Light --exact
Found OpenSSL.Light [ShiningLight.OpenSSL.Light] Version 4.0.2
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
This package requires the following dependencies:
- Packages
  Microsoft.VCRedist.2015+.x64
Downloading https://slproweb.com/download/Win64OpenSSL_Light-4_0_2.msi
5.73 MB / 5.73 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\url-mia-2026>
History restored
PS C:\url-mia-2026> Get-ChildItem "C:\Program Files\OpenSSL-Win64\bin\openssl.exe"
```

```
PS C:\url-mia-2026> Get-ChildItem "C:\Program Files\OpenSSL-win64\bin\openssl.exe"

Directory: C:\Program Files\OpenSSL-win64\bin

Mode                LastWriteTime         Length Name
----                -
-a----           2026-08-25   9:21 AM             821760 openssl.exe

PS C:\url-mia-2026> $opensslPath = "C:\Program Files\OpenSSL-win64\bin"
>> [Environment]::SetEnvironmentVariable("Path", $env:Path + ";" + $opensslPath, "user")
>> $env:Path += ";C:\Program Files\OpenSSL-win64\bin"
PS C:\url-mia-2026> $env:Path += ";C:\Program Files\OpenSSL-win64\bin"
PS C:\url-mia-2026> openssl enc -aes-256-cbc -salt -pbkdf2 -iter 100000 -in .\mensaje_DiegoApellido.json -out .\mensaje_DiegoLicardie.enc
Can't open ".\mensaje_DiegoApellido.json" for reading, No such file or directory
6C750000:error:80000002:system library: BIO_new_file: No such file or directory:crypto\bio\bss_file.c:73:calling fopen(.
.\mensaje_DiegoApellido.json, rb)
6C750000:error:10000000:
PS C:\url-mia-2026> openssl enc -aes-256-cbc -salt -pbkdf2 -iter 100000 -in .\mensaje_DiegoLicardie.json -out .\mensaje_DiegoLicardie.enc
Can't open ".\mensaje_DiegoLicardie.json" for reading, No such file or directory
24670000:error:80000002:system library: BIO_new_file: No such file or directory:crypto\bio\bss_file.c:73:calling fopen(.
.\mensaje_DiegoLicardie.json, rb)
24670000:error:10000000: BIO routines: BIO_new_file: no such file:crypto\bio\bss_file.c:81:
PS C:\url-mia-2026> cd semana_8\lab8
PS C:\url-mia-2026\semana_8\lab8> openssl enc -aes-256-cbc -salt -pbkdf2 -iter 100000 -in .\mensaje_DiegoLicardie.json -out .\mensaje_DiegoLicardie.enc
enter AES-256-CBC encryption password:

Verifying - enter AES-256-CBC encryption password:

PS C:\url-mia-2026\semana_8\lab8> █
```

```
PS C:\url-mia-2026\semana_8\lab8> Get-ChildItem

Directory: C:\url-mia-2026\semana_8\lab8

Mode                LastWriteTime         Length Name
----                -
-a----           2026-08-28   8:47 AM             176 mensaje_DiegoLicardie.enc
-a----           2026-08-28   8:40 AM             153 mensaje_DiegoLicardie.json

PS C:\url-mia-2026\semana_8\lab8> Get-Content .\mensaje_DiegoLicardie.enc
Salted__':8}+0,0WRB5Y4`qb
ËCXEj3ÄA\4"AP!>E=äö0·%0! 'tP*(-Typ,ÄÄsE „R;Äq0]E5,90>>»HkÄ
&uß»öSé/YéÜj4**20%$-Që0·EÖE"D765Z
PS C:\url-mia-2026\semana_8\lab8> openssl enc -d -aes-256-cbc -pbkdf2 -iter 100000 -in .\mensaje_DiegoLicardie.enc -out .\mensaje_DiegoLicardie_recuperado.json
enter AES-256-CBC decryption password:

PS C:\url-mia-2026\semana_8\lab8> openssl dgst -sha256 -hmac "www2007" .\mensaje_DiegoLicardie.json
HMAC-SHA2-256(.\mensaje_DiegoLicardie.json)= 71fa404cc2855ad9ff53c0f04abd90cd36dd1588f03c05f417acff15aebdebd7
PS C:\url-mia-2026\semana_8\lab8> █
```

¿Cuál es el HMAC-SHA256 obtenido?

HMAC-SHA2-256(.mensaje\_DiegoLicardie.json)=

71fa404cc2855ad9ff53c0f04abd90cd36dd1588f03c05f417acff15aebdebd7

## Preguntas

- 1.) Estoy protegiendo la información del archivo
- 2.) Si subo la contraseña cualquiera que tenga el link de mi repositorio puede verlo y acceder al archivo encriptado
- 3.) SHA-256 solo verifica integridad y HMAC-SHA256 combina el hash con una clave secreta, así que además verifica **la autenticidad**.